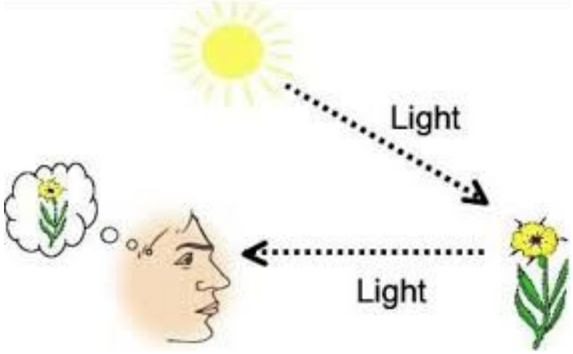
Computer Graphics

Unit 4 Light, Colour, Shading & Hidden Surfaces

Illumination Models and Shading-

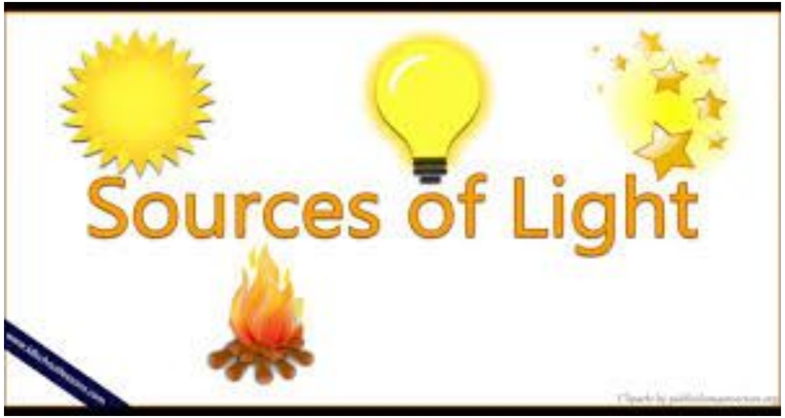
- Illumination- Lighting, Brightness
- Shading- Dark portion, light portion
- Intensity- Quantity of Illumination
- We see the object because object reflect light into our eyes.
- In this unit we concentrate on the location and qualities of the light that falls on the objects and the way in which objects interacts with it.
- A model for the interaction of light with a surface is called an illumination model.
  Illumination model is used to **calculate the intensity of light** that is reflected a given point on surface.



Sources of Light-

Many objects are available which produce or emit light.

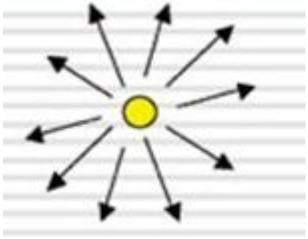
Exa. Sun, Lamp, stars, candle, bonfire (i.e Light emitting source)



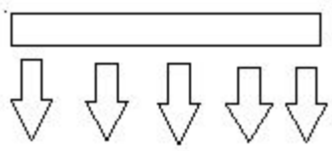
Sources of Light...

Light emitting sources divided into 2 types

1) **Point Sources-** Exa.Sun, Candle

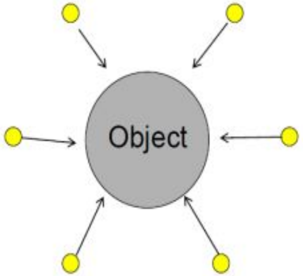


2) **Distributed Sources**- Exa. Tubelight



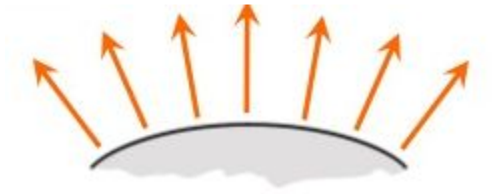
Ambient Light-

- All objects which are not emitting light are grouped under separate category called **Light reflecting sources.**
- Light from environment
- The surface that is not exposed to direct light may still be lit up by reflections from other nearby objects.
- This light which is reflected from somewhere is called as **ambient light(background light)**.



Diffuse Illumination(Reflection)-

- When a light falls on any surface it can be absorbed by the surface, while the rest will be reflected randomly in all direction.
- In diffuse illumination , incoming light is not reflected in a single direction but is scattered almost in all direction.



Diffuse Illumination...

- The ratio of light reflected from the surface to the total incoming light is called **coefficient of reflection or reflectivity** (K_d)**.**
- The white surface is having its reflectivity close to 1, hence it reflects almost all incoming light.
- In case of black surface absorbs most of the incoming light since its reflectivity is near to 0.
- So the value of reflectivity (K_d) is assigned in the range of 0 to 1,depending on the property or nature of surface.

Diffuse Illumination...

Intensity of the diffuse reflection at any point is,

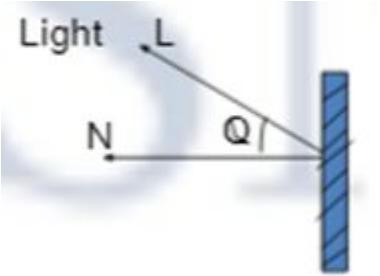
$$I = K_d \cdot I_a$$

Where, I_a = Intensity of ambient light

K_d = Coefficient of reflection

Point source Illumination-

- In case of point source,the light orientation from a fixed place and it comes from a particular direction.
- Point source produces rays/light from single point.Exa.bulbs,candle.
- Surface which are near the light source & facing towards light will receive more light.
- The illumination decreased by a factor of cosθ
  Where θ is the angle of incidence between
  L Light & N Normal to plane



- cosθ is range from 0 to 1
- The amount of light depends upon cosθ
- Surface will be illuminated by point source
  cosθ= N. L

Point source Illumination..

The Diffuse reflection will be,

$$I = K_d \,.\, I_p \,.\, \cos\theta$$

Where, I_p = Intensity point source light

K_d = Coefficient of reflection

θ = angle of incidence

We can write

$$I = K_d \,.\, I_p \,.\, (N.\, L)$$

Specular Reflection-

- The light is reflected in a single direction.
- Exa. Metal surfaces, Mirrors, shiny plastics, glass, gold silver coated surfaces etc
- Specular reflection is independent of color.
- In specular reflection light comes in, strikes the surface and then bounces back.

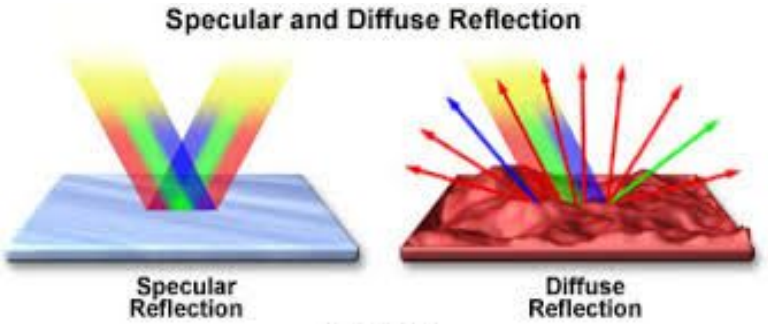
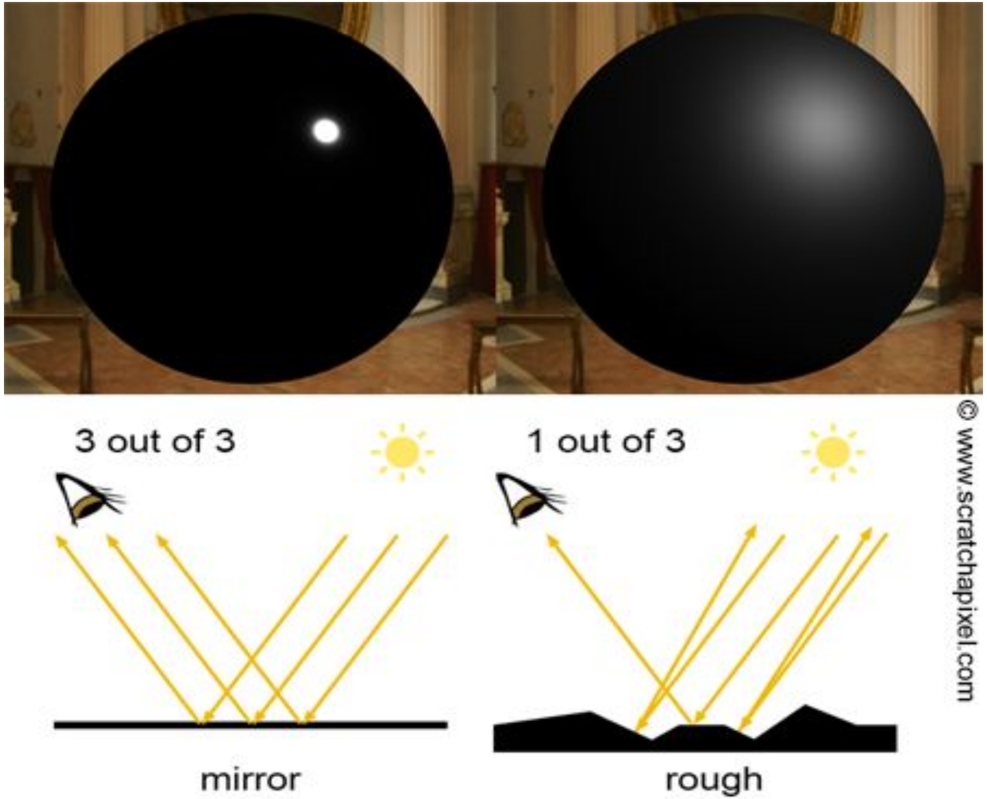


Figure 1

- The specular light is a white highlight reflection seen on smooth shiny objects.
- Depends on both surface orientation(like shiny, glossy) & viewer location.



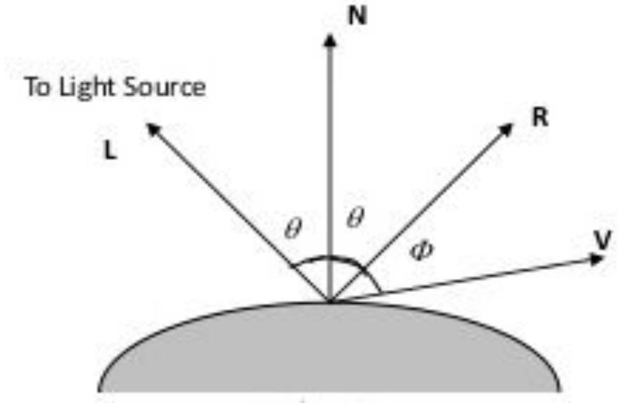
Specular Reflection...

From Fig L = direction of light source, N= surface normal, R= direction of reflected light, V= Viewer. The angle which is the reflected beam makes with surface normal is called angle of reflection. Angle Φ is the viewing angle relative to the specular reflection direction R. For ideal reflector Φ should be zero.

$I_{sp} = I \cdot K_s \cdot (\cos \Phi)^n$ Where I= Intensity of light source, K_s = specular reflectivity.

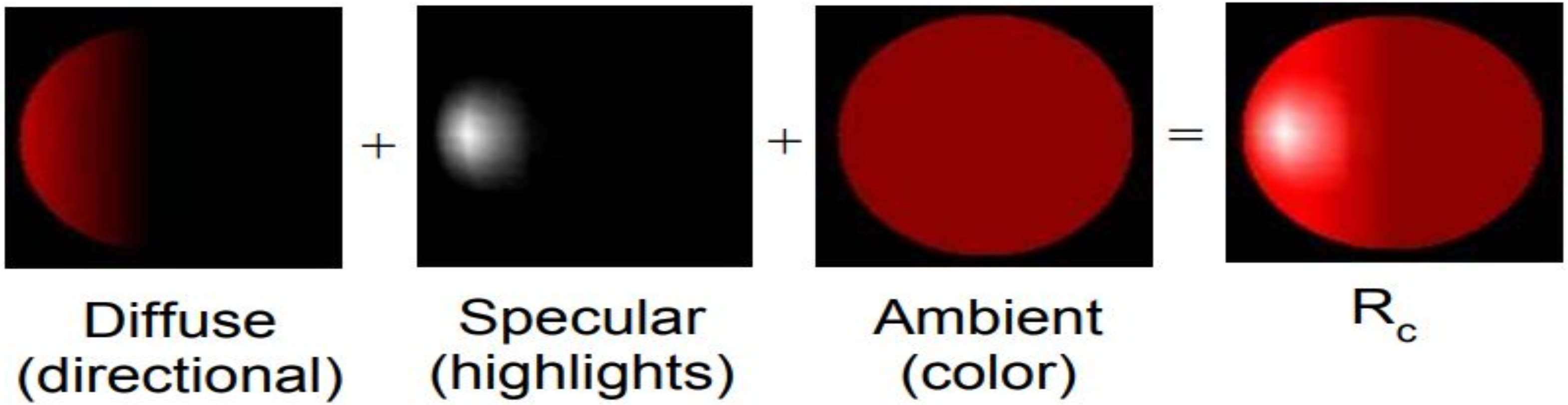
But V & R are unit vectors in the viewing & specular reflection we can write

$I_{sp} = I \cdot K_s \cdot (V \cdot R)^n$



Phong model-

- Simple 3 parameter model
  - The sum of 3 illumination terms:
    - **Diffuse :** non-shiny illumination and shadows
    - **Specular :** bright, shiny reflections
    - **Ambient :** 'background' illumination



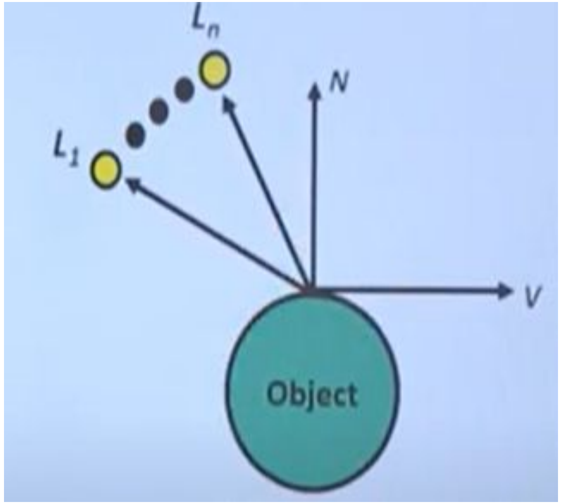
Combined Diffuse and Specular Reflections with Multiple Light source-

We can combined Diffuse and specular reflection for a single point light source from a point on an illumination surface as

$$I = I_{diff} + I_{spec}$$

$$= K_d \, . \, I_a + K_d \, . \, I_p \, . \, (N. L) + I \, . \, K_s \, . \, (V. R)^n$$

If we consider multiple light source then by using normalization the equation will be,



$$I = K_d \, . \, I_a + \sum_{i=1}^{n} I_{pi} \left[K_d \, . \, (N. L_i) + K_s \, . \, (V. R_i)^n \right]$$

Warn Model-

- The Warn model provides a method for **simulating studio lighting effects by controlling light intensity in different directions.**
- It provides a way model Studio light with varying intensity in different directions
- Light sources are modeled as points on a reflecting surface, using the Phong model for the surface points.
- Then the intensity in different directions is controlled by selecting values for the **Phong exponent**
- In addition, light controls and spotlighting, used by studio photographers can be simulated in the Warn model.
- Flaps are used to control the amount of light emitted by a source In various directions

Properties of Light

- A light source produced by sun or electric bulb emits all frequencies within the visible range to give white light.
- When this light is incident upon an object, some frequencies are absorbed and some are reflected by the object.
- The combination of reflected frequencies decides the color of the object.

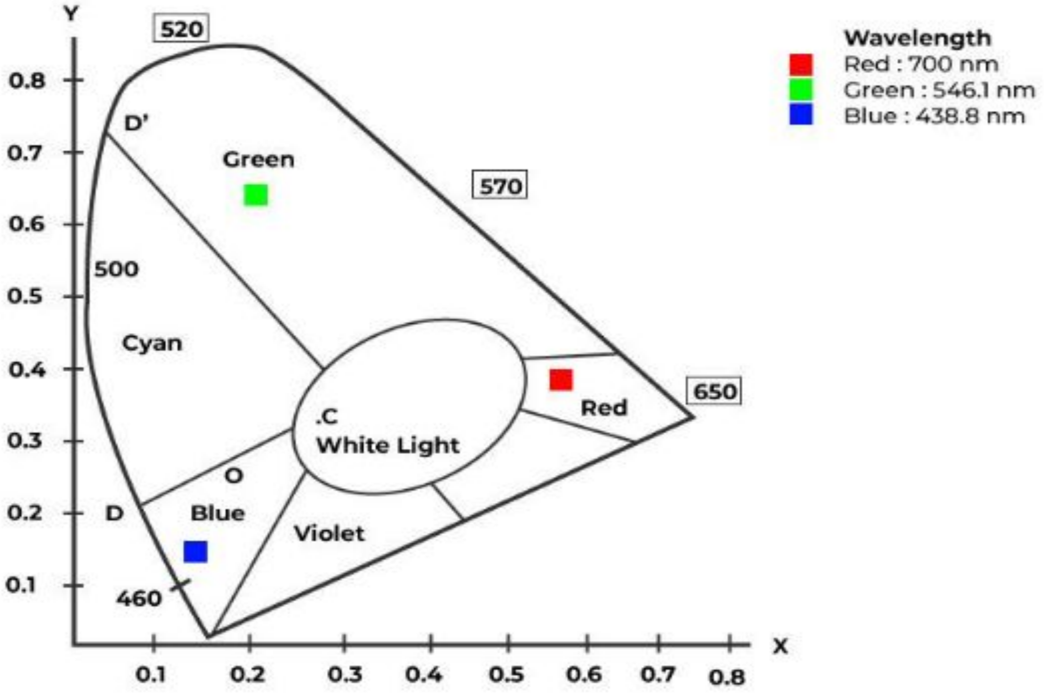
Properties of Light

- If the lower frequencies are predominant in reflected frequencies, the object color is red .
- In this case, we can say that the perceived light has a dominant frequency at the red end of the spectrum.
- The dominant frequency decides the color of the object. Due to this reason dominant frequency is also called the hue or simply the color.

Properties of Light

- There are two more properties Brightness and saturation (purity)
- The brightness refers to the intensity of the perceived light.
- The saturation describes the purity of the color.
- When the two properties purity and dominant frequency are used collectively to describe the color characteristics are referred to as chromaticity.

CIE Chromaticity Diagram-



CIE Chromaticity Diagram..

- The chromaticity diagram represents the spectral colours and their mixtures based on the values of the primary colours(i.e Red, Green, Blue) contained by them.
- Chromaticity contains two parameters i.e, hue and saturation. When we put hue and saturation together then it is known as Chrominance
- Chromaticity diagram represents visible colours using X and Y as horizontal and vertical axis.
- The various saturated pure spectral colours are represented along the perimeter of the curve representing the three primary colours – red, green and blue.
- Point C marked in chromaticity diagram represents a particular white light formed by combining colours having wavelength :RED: 700 nm , GREEN : 546.1 nm,BLUE: 438.8 nm
- In Chromaticity diagram colours on boundary are completely saturated. The corners in this chromaticity diagram represents by three primary colours (Red, Green and Blue).

CIE Chromaticity Diagram..

Uses of C.I.E Chromaticity Diagram in Computer Graphics:

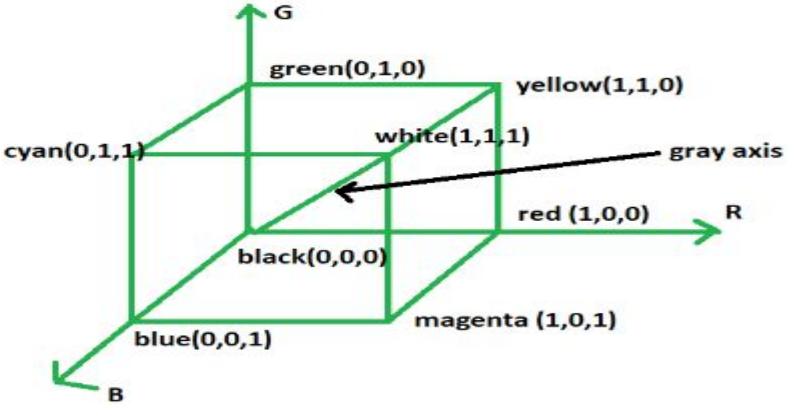
- It is used to evaluate a color against a gamut(range).
- It is used to represent all chromaticities visible to the average person.
- It is used to find the range of colors.
- It is used to find dominant and complementary colors.

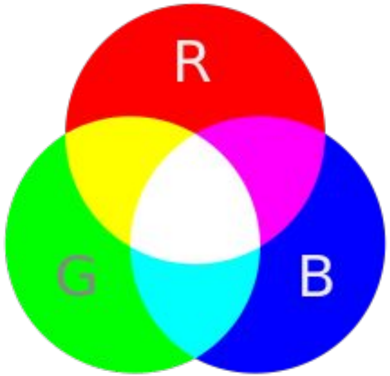
Advantages:

- Simple geometric construction for mixing two or more colours.
- Both Perceptual attributes used: hue and saturation.

RGB-

- The RGB color model is one of the most widely used color representation method in computer graphics. It use a color coordinate system with three primary colors: R(red), G(green), B(blue)
- Each primary color can take an intensity value ranging from 0(lowest) to 1(highest). Mixing these three primary colors at different intensity levels produces a variety of colors. The collection of all the colors obtained by such a linear combination of red, green and blue forms the cube shaped RGB color space.





RGB..

The corner of RGB color cube that is at the origin of the coordinate system corresponds to black, whereas the corner of the cube that is diagonally opposite to the origin represents white. The diagonal line connecting black and white corresponds to all the gray colors between black and white, which is also known as **gray axis**.

In the RGB color model, an arbitrary color within the cubic color space can be specified by its color coordinates: (r, g, b).

Example:

```
(0, 0, 0) for black, (1, 1, 1) for white,(1, 1, 0) for yellow
```

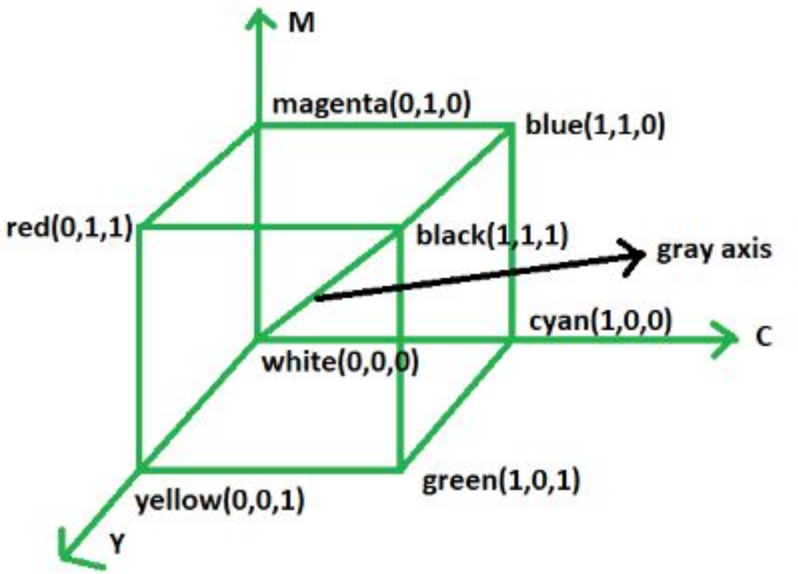
Color specification using the RGB model is an **additive process**. We begin with black and add on the appropriate primary components to yield a desired color.

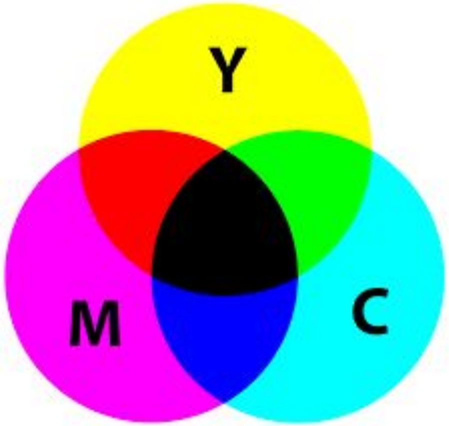
The concept RGB color model is used in Display monitor,TV, Camera, Web graphics

CMY-

The CMY color model use a **subtraction process** and this concept is used in the **printer**.

In CMY model, we begin with white and take away the appropriate primary components to yield a desired color.





CMY..

Example:

If we subtract red from white, what remains consists of green and blue which is cyan. The coordinate system of CMY model use the three primaries' complementary colors:

```
C(cray), M(magenta) and Y(yellow)
```

The corner of the CMY color cube that is at (0, 0, 0) corresponds to white, whereas the corner of the cube that is at (1, 1, 1) represents black. The following formulas summarize the conversion between the two color models:

$$\begin{bmatrix} R \\ G \\ B \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} C \\ M \\ Y \end{bmatrix} \qquad \begin{bmatrix} C \\ M \\ Y \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

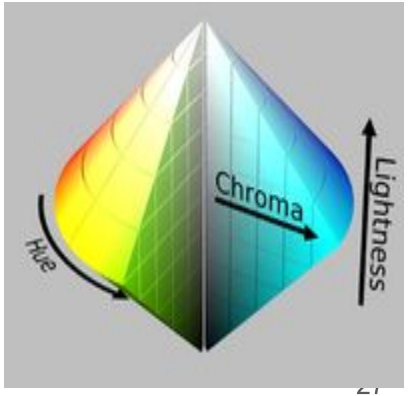
HSV-

- An HSV color model is the most accurate color model as long as the way humans perceive colors. How humans perceive colors is not like how RGB or CMY make colors.
- They are just primary colors fused to create the spectrum. The H stands for Hue, S stands for Saturation, and the V stand for value.
- Imagine a cone with a spectrum of red to blue from left to right, and from the centre to the edge, the color intensity increases. From bottom to up, the brightness increases.
- Hence resulting white at the center

  up layer.
- A pictographic representation is shown

  in figure

Angle (in degree)	Color
0-60	Red
60-120	Yellow
120-180	Green
180-240	Cyan
240-300	Blue
300-360	Magenta



HSV..

Hue: Hue tells the angle to look at the cylindrical disk. The hue represents the color. The hue value ranges from o to 360 degrees.

Saturation: The saturation value tells us how much quantity of respective color must be added. A 100% saturation means that complete pure color is added, while a 0% saturation means no color is added, resulting in grayscale.

Value: The value represents the brightness concerning the saturation of the color. the value 0 represents total black darkness, while the value 100 will mean a full brightness and depend on the saturation.

HSV..

Advantages:

The advantage of HSV is that it generalizes how humans perceive color. Hence it is the most accurate depiction of how we feel colors on the computer screen

Applications:

- HSV model is used in histogram equalization.
- Converting grayscale images to RGB color images.
- HSV used in color pickers in image editing software.

Shading Algorithms-

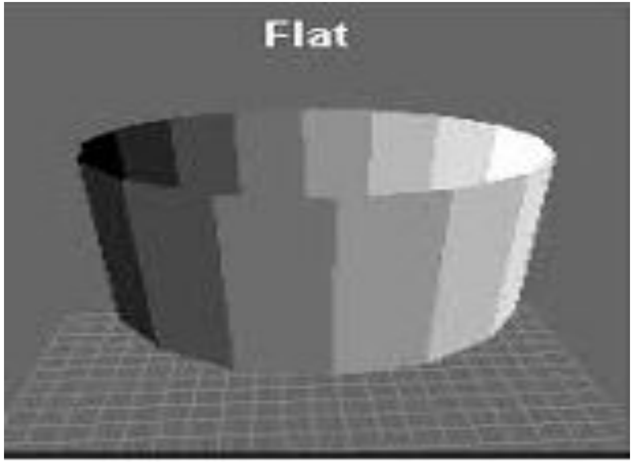
- The Process of assigning a color to the pixel.
- Shading simply defined as the smooth transition from one color to next.

Divided into 4 types of Shading Algorithms

1) Constant Shading(Flat Shading)- Lighting Computation per polygon.
2) Gouraud Shading - Lighting Computation per vertex.
3) Phong Shading - Lighting Computation per pixel.
4) Halftone Shading -

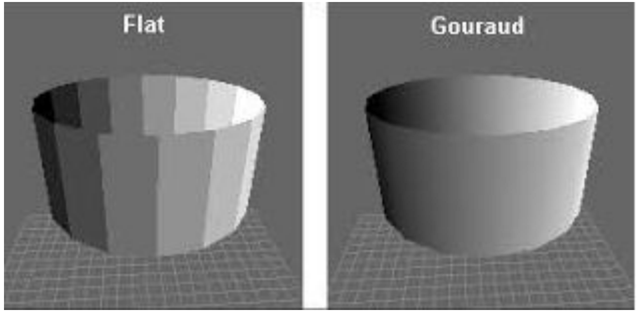
1) Constant Shading(Flat Shading)-

- Easy method
- In this computation made for each surface/polygon.
- In this shading method, for each polygon, a **single intensity** is calculated.
- Then with the same intensity value all points on that surface of the polygon are displayed.



2) Gouraud Shading-

- It is the example of intensity interpolation of method.
- Here polygon surfaces are rendered by linearly interpolating values.
- This method removes intensity discontinuities between adjacent planes of surface representation by linearly varying the intensity over each plane.
- Here intensity values are match at plane boundaries.

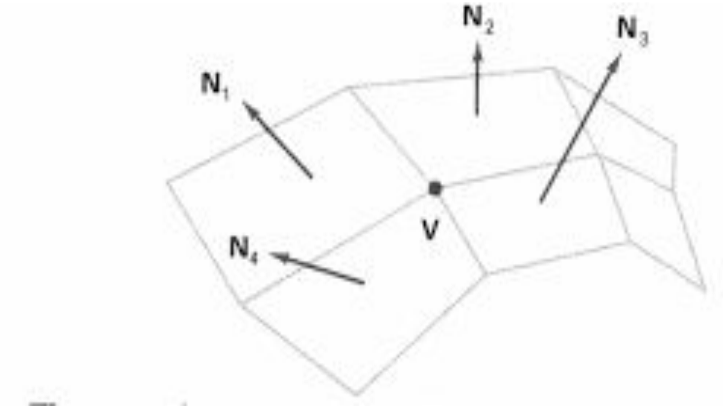


2) Gouraud Shading...

- In this shading all polygon surfaces are accessed by performing calculations:

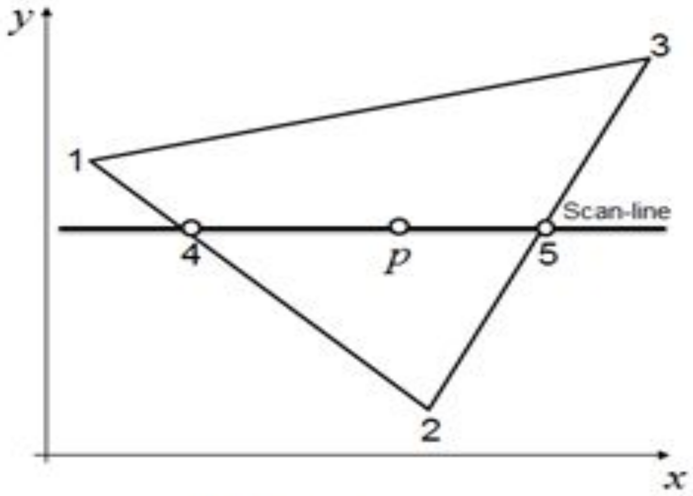
1) At each vertex of polygon, determine average unit normal vector.
2) Calculate vertex intensity.
3) Linearly interpolate that vertex intensity over the surface of polygon.

- 1) To determine a normal vector N at vertex V in fig we use average normal vector is $N = N_1 + N_2 + N_3 + N_4$ where N is then normalized by dividing it by 4.



2) Gouraud Shading...

- 2) By applying Illumination we get intensity of each vertex.
- 3) Now we know all vertex intensity then we can find all other intensities for the surface(Intensity interpolation).



Figure

$$I_4 = \frac{y_4 - y_2}{y_1 - y_2} I_1 + \frac{y_1 - y_4}{y_1 - y_2} I_2$$

$$I_5 = \frac{y_5 - y_2}{y_3 - y_2} I_3 + \frac{y_3 - y_5}{y_3 - y_2} I_2$$

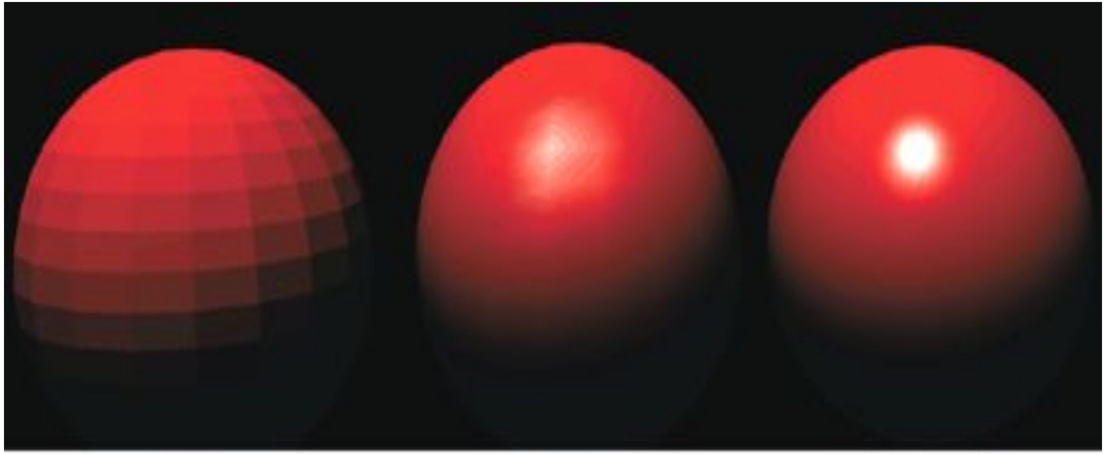
$$I_p = \frac{x_5 - x_p}{x_5 - x_4} I_4 + \frac{x_p - x_4}{x_5 - x_4} I_5$$

2) Gouraud Shading...

- This process is repeated for each scan line passing through polygon. We get all intensity to fill the polygon.
- **Advantages-**
- The discontinuity in intensity is removed as compared to constant shading.
- **Disadvantages-**
- Sharp drop of intensity values on the polygon surface cannot be displayed.
- Bright & Dark streaks appearing on the surface known as mach bands.

3) Phong Shading-

- This method is more accurate also involves approximation of the surface normal at each point along a scan line , then calculating intensity using approximated normal vector at that point, displaying more realistic highlights on the surface.



Flat Gouraud Phong

3) Phong Shading...

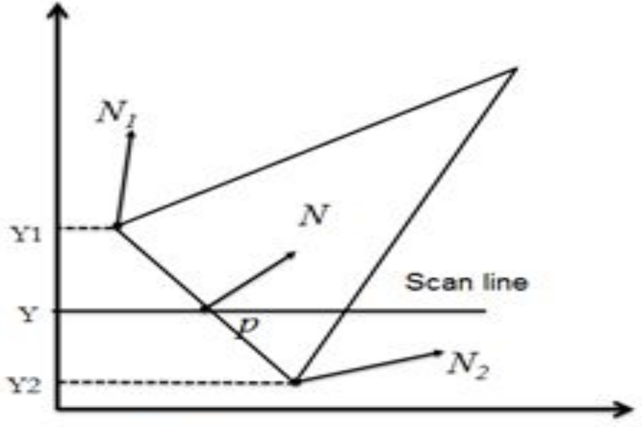
- Steps of Phong modelling(Normal vector interpolation)-

1) At each vertex of polygon, determine average unit normal vector.
2) Linearly interpolate the vertex normals.
3) Calculate the pixel intensity of each scan line.

- This method interpolate the normal vectors at the bounding points along the scan line, see in fig

3) Phong Shading...

To find normal vector N at a point p in triangle , we find N by interpolation and after finding N we can find all normal vector on scan line Y. After that we can find intensity of each pixel.



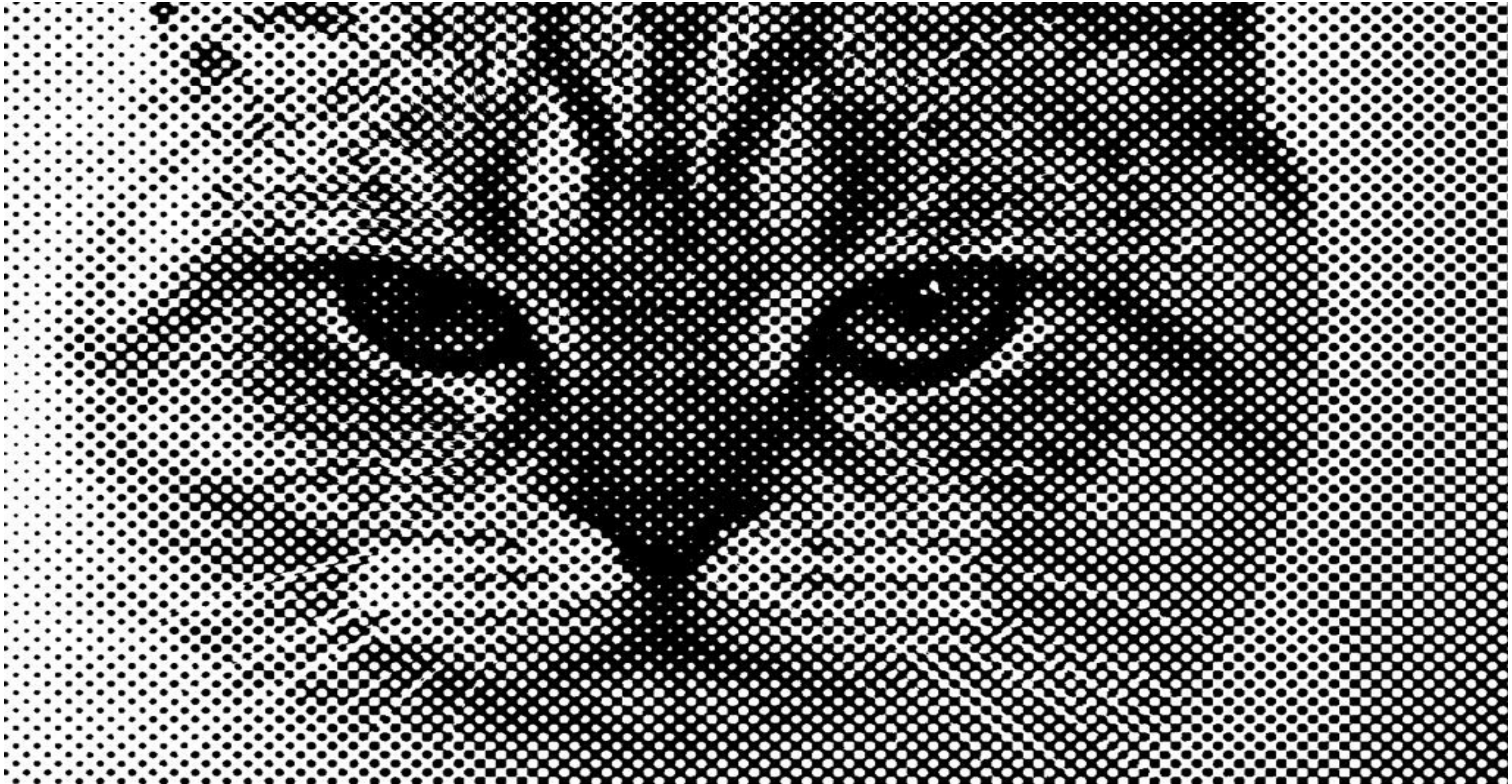
Figure

$$N = \frac{y - y_2}{y_1 - y_2} N_1 + \frac{y_1 - y}{y_1 - y_2} N_2$$

3) Phong Shading...

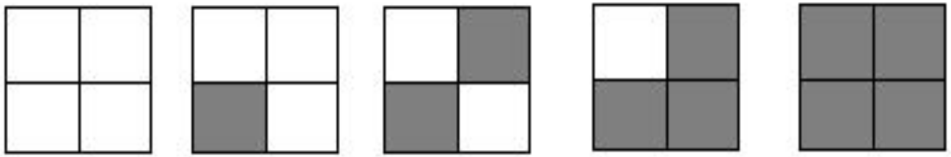
- Displays more highlight (specular) on the surface
- **Advantages-**
- More accurate result
- More realistic highlights.
- **Disadvantages-**
- More Time consuming i.e slower
- Require more calculation.
- Cost of shading increases.

4) Halftone Shading-



4) Halftone Shading...

- Many times require less number of color intensity range.
- The halftoning is commonly used in printing black and white photographs in newspapers, magazines, and books.
- A halftoning method is used to increase the number available intensities by incorporating multiple pixel positions into the display of each intensity value.
- The phenomenon of apparent increase in the number of available intensities by considering combine intensity of multiple pixels is known as halftoning.
- Each intensity position in the original scene is replaced with a rectangular pixel grid.

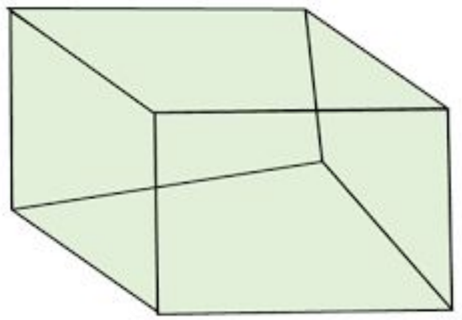


4) Halftone Shading...

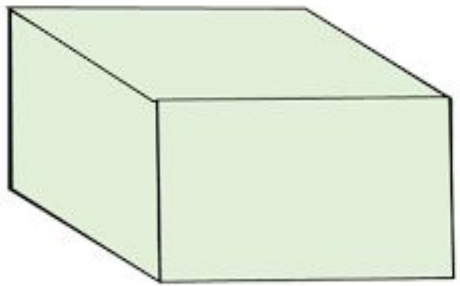
- For any n x n pixels for each grid on bilevel system $n^2 + 1$ intensity levels can be generated.
- I.e. for 3 X 3 pixel grid on a bilevel system ,10 intensity levels be obtained.

Hidden Surfaces-

- Hidden surfaces of objects inside the viewing object that should not be seen.
- To determine which lines or surfaces of the objects are visible, either from center of projection or along the direction of projection, so we can display only the visible lines or surface this process is known as visible surface detection or hidden surface elimination.



Object with hidden line



Object when hidden lines removed

Back-Face Detection

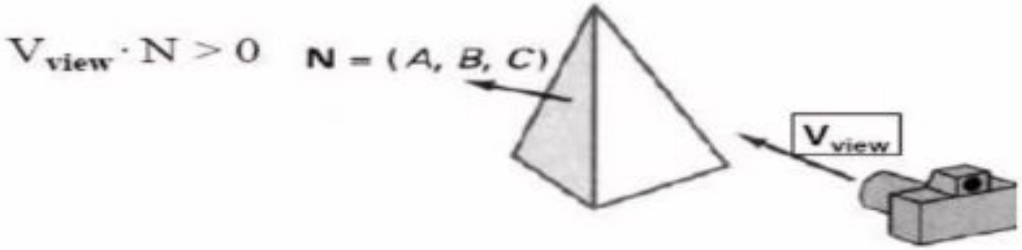
- When we project 3-D objects on a 2-D screen, we need to detect the faces that are hidden on 2D.
- **Back-Face detection**, also known as **Plane Equation method,** is an object space method in which objects and parts of objects are compared to find out the visible surfaces.
- Let us consider a triangular surface that whose visibility needs to decided. The idea is to check if the triangle will be facing away from the viewer or not. If it does so, discard it for the current frame and move onto the next one.
- Each surface has a normal vector. If this normal vector is pointing in the direction of the center of projection, then it is a front face and can be seen by the viewer. If this normal vector is pointing away from the center of projection, then it is a back face and can not be seen by the viewer.

- A fast and simple object-space method for locating back faces
- A point (x,y,z) is "inside" a polygon surface with plane parameters A, B, C, D if :

  $$Ax + By + Cz + D < 0$$
- When an inside point is along the line of sight to the surface, the polygon must be a back face and so cannot be seen.

Back-Face Detection

- The test is simplified by considering the normal vector N to the polygon and the viewing vector V

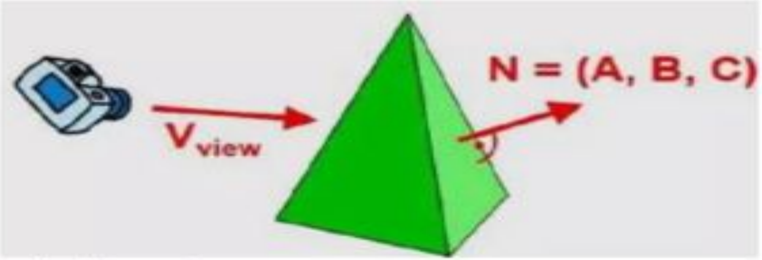


- This is back face if

  $V \cdot N > 0$
- A polygon is a back face if $V_{view}.N > 0$. But it should be kept in mind that after application of the viewing transformation, viewer is looking down the negative Z-axis. Therefore, a polygon is back face if :
- $(0, 0, -1).N > 0$ or if $C < 0$

Back–Face Detection

- In a right–handed viewing system with viewing direction along the negative Z_v axis, the polygon is a back face if $C < 0$. Also, we cannot see any face whose normal has z component $C = 0$, since your viewing direction is towards that polygon. Thus, in general, we can label any polygon as a back face if its normal vector has a z component value –
- **$C <= 0$**



- **Algorithm for right–handed system :**

1) Compute N for every face of object.
2) If (C.(Z component) < 0)
        then a back face and don't draw
   else
        front face and draw

Thus, for the right–handed system, if the Z component of the normal vector is negative, then it is a back face. If the Z component of the vector is positive, then it is a front face.

Back-Face Detection

- Similar methods can be used in packages that employ a left-handed viewing system. In these packages, plane parameters A, B, C and D can be calculated from polygon vertex coordinates specified in a clockwise direction (unlike the counterclockwise direction used in a right-handed system).
- Also, back faces have normal vectors that point away from the viewing position and are identified by C >= 0 when the viewing direction is along the positive *Zv*
- axis. By examining parameter C for the different planes defining an object, we can immediately identify all the back faces.
- the **Algorithm for left-handed system** :

1) Compute N for every face of object.
2) If (C.(Z component) > 0)

    then a back face and don't draw

  Else

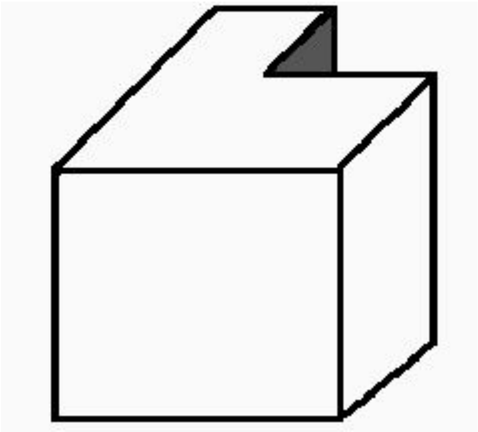
    front face and draw

The Back-face detection method is very simple. For the left-handed system, if the Z component of the normal vector is positive, then it is a back face. If the Z component of the vector is negative, then it is a front face.

Back-Face Detection

Limitations on Back-face removal algorithm

1. It can only be used on solid objects modeled as a polygon mesh. This is the most general modeling construct for scan line graphics systems. Even if objects are defined in a different manner, e.g. by parametric cubic patches or implicit equations, the renderer usually converts everything to polygons to render.
2. It works fine for convex polyhedra but not necessarily for concave polyhedra as shown below in the example of a partially hidden face, that will not be eliminated by Back-face removal.

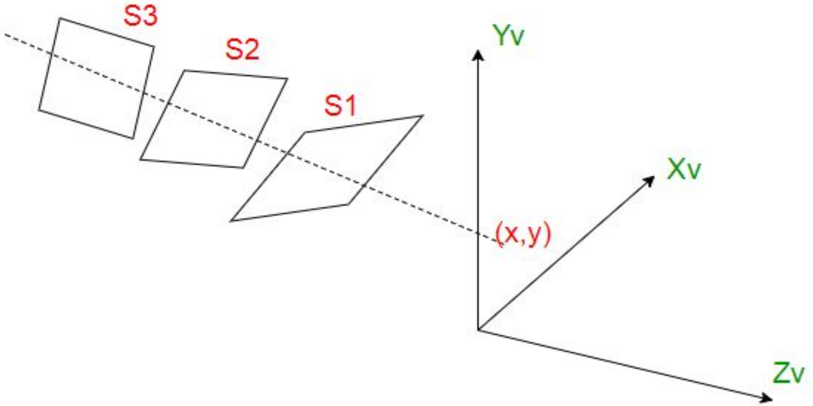


Depth buffer (z) Algorithm-

- When viewing a picture containing non transparent objects and surfaces, it is not possible to see those objects from view which are behind from the objects closer to eye. To get the realistic screen image, removal of these hidden surfaces is must. The identification and removal of these surfaces is called as the **Hidden-surface problem.**
- Z-buffer, which is also known as the Depth-buffer method is one of the commonly used method for hidden surface detection. It is an **Image space method**. Image space methods are based on the pixel to be drawn on 2D. For these methods, the running time complexity is the number of pixels times number of objects. And
- space complexity is two times the number of pixels because two arrays of pixels are required, one for **frame buffer** and the other for the **depth buffer**.
- The Z-buffer method compares surface depths at each pixel position on the projection plane. Normally z-axis is represented as the depth.

Depth buffer (z) Algorithm…

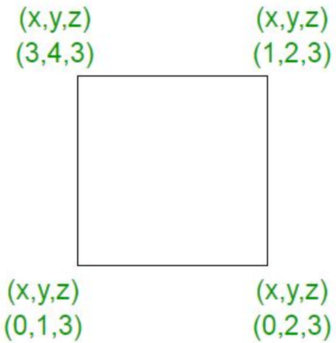
```
First of all, initialize the depth of each pixel.
i.e,  d(i, j) = infinite (max length)
Initialize the color value for each pixel
as c(i, j) = background color
for each polygon, do the following steps :

for (each pixel in polygon's projection)
{
    find depth i.e, z of polygon
    at (x, y) corresponding to pixel (i, j)

    if (z < d(i, j))
    {
        d(i, j) = z;
        c(i, j) = color;
    }
  }
```



Depth buffer (z) Algorithm…

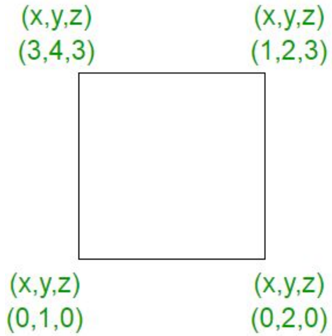
- In this method, 2 buffers are used :

1. Frame buffer
2. Depth buffer



∞	∞	∞
∞	∞	∞
∞	∞	∞
∞	∞	∞

3	3	3
3	3	3
3	3	3
3	3	3



∞	∞	∞
∞	∞	∞
∞	∞	∞
∞	∞	∞

3	3	3
2	2	2
1	1	1
0	0	0

Depth buffer (z) Algorithm…

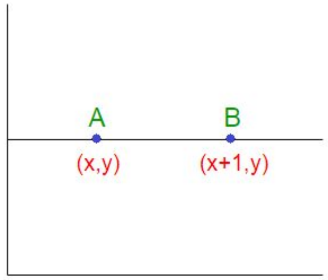
- In this method, each surface is processed separately one position at a time across the surface. After that the depth values i.e, the z values for a pixel are compared and the closest i.e, (smallest z) surface determines the color to be displayed in frame buffer. The z values, i.e, the depth values are usually normalized to the range [0, 1]. When the z = 0, it is known as ***Back Clipping Pane*** and when z = 1, it is called as the ***Front Clipping Pane***.
- **Calculation of depth :**

  As we know that the equation of the plane is :

  `ax + by + cz + d = 0, this implies`

  `z = -(ax + by + d)/c, c!=0`

  Calculation of each depth could be very expensive, but the computation can be reduced to a single add per pixel by using an increment method as shown in figure :



Depth buffer (z) Algorithm…

Let's denote the depth at point A as Z and at point B as Z'. Therefore :

```
AX + BY + CZ + D = 0 implies
Z = (-AX - BY - D)/C  ------------(1)
Similarly, Z' = (-A(X + 1) - BY -D)/C   ----------(2)
Hence from (1) and (2), we conclude :
Z' = Z - A/C  ------------(3)
```

Hence, calculation of depth can be done by recording the plane equation of each polygon in the (normalized) viewing coordinate system and then using the incremental method to find the depth Z.
So, to summarize, it can be said that this approach compares surface depths at each pixel position on the projection plane.
Object depth is usually measured from the view plane along the z-axis of a viewing system.

Disadvantage-Cannot be applied on transparent surfaces i.e, it only deals with opaque surfaces.